



Written Expression in Individuals with Autism Spectrum Disorder: A Meta-Analysis

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Abstract

Although studies exist measuring the effectiveness of writing interventions for students with autism spectrum disorder (ASD), research assessing the writing skills for this group is sparse. The present study identified differences in the written expression of individuals with ASD compared to typically developing (TD) peers, using variables selected from 13 different studies. Using Pearson Product Moment-correlation the relationship between the quality of research studies and the magnitude of the effect sizes was examined. Findings indicate significant differences in the following components of written expression; length, legibility, handwriting size, speed, spelling, and overall structure, highlighting the need for future research to determine if the characteristics of written expression in individuals with ASD are similar to other struggling writers.

Keywords Autism spectrum disorder · Writing · Composition · Meta-analysis · Autism

Introduction

Written expression is one of the primary means through which students demonstrate their knowledge about a content area. Students in today's schools are required to write essays, short concise responses on assessments, and college applications. With the increase of digital technologies, students must also be prepared to share thoughts in writing at work. Since all students in the United States, including students with autism spectrum disorder (ASD), are included in the statewide assessment and accountability systems (which includes assessing students on whether or not they have met standards in literacy), educators have become increasingly concerned with identifying and meeting the unique educational needs of these students. The concerns are commensurate with the rise in the number of students receiving special educational services under the category of autism, which tripled in the 10 years between 2003 and 2012 (U.S. Department of Education 2013). Although a number of studies measure the effectiveness of writing interventions for

students with ASD (Asaro-Saddler 2016; Pennington and Delano 2012), research on writing skills and the development of writing skills for this group is sparse.

ASD is a condition which impacts an individual's social communication and interaction (American Psychiatric Association 2013). Individuals with ASD may exhibit deficits in understanding social-emotional reciprocity, verbal and non-verbal communication, and human relationships. In addition, individuals with ASD display restricted, repetitive patterns of behavior which may manifest themselves as idiosyncratic verbal patterns, and insistence on sameness or a fixation on particular interest. Individuals with ASD may have hyper- or hypo-reactivity to sensory input. These characteristics may impact an individual's academic progress and achievement. The IQ scores for individuals with ASD span across normal distribution, but show variability and inconsistency among subtests (Griswold et al. 2002; Mayes and Calhoun 2003). Rote memory skills appear to be relatively intact, which may explain high scores in vocabulary in some individuals with ASD (Barnhill et al. 2000). However, research shows deficits in reasoning, abstract thinking, and complex language use (Meyer 2002). Many individuals with ASD experience difficulties with reading comprehension, written expression, and graphomotor skills (Mayes and Calhoun 2003; Whitby and Mancil 2009). It is important to determine if the difficulties individuals with ASD have with writing are unique

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to diagnosis, or are similar to difficulties faced by students with other disabilities or by their typically developing peers.

Elements of Written Expression

Writing is a highly complex activity through which people communicate to accomplish a number of goals including maintaining personal connections, influencing others, keeping records, and self-reflection (National Council of English Teachers [NCTE] 2016). Writing requires the coordination of multiple simultaneous demands, which involve the mechanics of writing (handwriting, spelling, sentence construction, typing), content (ideation, genre, vocabulary), and organization. Students develop strategies in writing to support comprehension of concepts within the curriculum, and use writing to express their knowledge and views (Graham et al. 2012). The current study compares the performance of individuals with ASD on components of writing with the performance of typical developing individuals.

Mechanics of Writing

Basic writing skills, often referred to as the mechanics of writing, include handwriting, spelling, sentence construction, and typing (Graham et al. 2012). The acquisition of these skills enables students to communicate their ideas in a variety of different formats such as test responses, emails, and essays. Transcription skills, i.e., handwriting and spelling compete for limited cognitive resources. If spelling and handwriting are taxing it can affect the quality of student compositions. Vice versa, if basic skills such as handwriting and spelling are automatic, then resources in working memory are freed for higher level processes such as planning, generating language, and revising needed in written expression (Berninger et al. 1997). Although considered basic, these skills each have complexities and challenges of their own.

Handwriting “Handwriting is a perceptual-motor skill encompassing a blend of visual-motor coordination abilities, motor planning, cognitive, and perceptual skills, as well as tactile and kinesthetic sensitivities” (Feder and Majnemer 2007, p. 313). Despite the increase of keyboarding, handwriting is a necessary life skill, essential for communication and everyday tasks. If handwriting is not automatic, the effort required to produce a written text can place a strain on other writing processes (Graham et al. 2000), resulting in a negative impact on academic success (Feder and Majnemer 2007).

Elements of handwriting that contribute to overall legibility include spacing between words and/or letters, as well as the formation (slant and size) of the letters (Graham et al. 2006). Handwriting fluency (legibility and

speed) increases as students get older, although the rate of improvement slows. By the time students have completed elementary school, the improvement in handwriting has plateaued (Graham et al. 1998). Boys are more likely to exhibit handwriting fluency problems than girls (Berninger et al. 1997; Graham et al. 1998), concerning since the ratio of boys to girls with ASD is approximately 5:1 (Centers for Disease Control and Prevention 2014).

The writing task and quantity of writing required can impact handwriting legibility (Feder and Majnemer 2007; Maldarelli et al. 2015). In a study conducted by Graham et al. (2006), participants were asked to complete three different tasks, each one placing increased cognitive demands on the writer: (a) writing the alphabet from memory, (b) copying a paragraph, and (c) composition. On average participants exhibited better spacing and alignment when copying versus composing. Those categorized as poor handwriters were more likely to produce letters that included extra strokes and to exhibit greater variability in letter placement. Instruction in handwriting is a factor in the quality of handwriting (Berninger et al. 1997). Without instruction, writers may develop habits that are detrimental to the overall quality of their written expression. Feder and Majnemer (2007) posit that motor planning is the best predictor of legibility, but suggest sensory awareness of fingers may have an impact on writing performance which may be a concern for individuals with ASD. If the physical act of handwriting is too demanding, students may fail to develop a self-regulatory approach to writing content or they may avoid writing altogether (Graham et al. 2008; Graham and Harris 2006).

Spelling Spelling is a transcription skill which involves placing letters in order to represent a word, so as to convey meaning. Spell-check functions on electronic devices can assist writers with correct spelling, composition, and proofreading (Berninger 1999). According to *triple word form theory*, three kinds of processing are needed for correct spelling: (a) phonological, (b) orthographic, and (c) morphological (Berninger et al. 2010). In the natural course of development, there is considerable variation in the rate children develop these processes, but patterns are evident. In the early stages, when children first learn to write their name, they appear to replicate a visual pattern or create a logographic representation of the word (Berninger 1999; McNeill et al. 2013). Over time, students recognize positional constraints of certain letter combinations, e.g., *ck*, and learn to spell new words through analogy (Bourassa and Treiman 2001). As students get older, phonemic and phonological awareness become predictors of spelling ability (Garcia et al. 2010). Omission of letters in blends, or phonologically plausible spelling errors are common. As students recognize letter units and ‘rimes’, orthographic

coding skills improve (Berninger 1999; Garcia et al. 2010; Berninger et al. 2010).

Morphological processing contributes to spelling by enabling students to generate new words by putting together previously learned morphemes. As morphological processing develops students are better able to use inflectional endings and derivational forms. This in turn may help with the understanding and selection of vocabulary. Being able to recognize the morphological structure of words allows students to store and retrieve information needed to spell more efficiently. Spelling can tax working memory resources, which can result in students being less likely to choose words they cannot spell (Graham and Santangelo 2014). Transcription skills compete with text generation for cognitive resources, so if transcription skills like handwriting and spelling are not automatized it can negatively impact other cognitive processes needed in written expression. It has been noted that individuals with ASD demonstrate a relative strength in rote learning, which may support spelling (Mayes and Calhoun 2006). It is not clear whether IQ scores, or co-morbidity with other disorders contribute to low spelling scores found in some individuals with ASD.

Sentence Construction

Syntax refers to how words are sequenced to form sentences. A sentence is often referred to as ‘a complete thought,’ but there is more to sentence construction than ‘putting thoughts into words.’ Translating thoughts into text requires careful selection of vocabulary, attention to syntax, and an understanding of the reader (Myhill 2009). In order for a writer to construct sentences, he or she must have syntactical knowledge which includes rules about word order, grammar, punctuation, and function (Saddler 2013).

Very young children entering school may write incomplete sentences or add unneeded words, possibly because they write the way they talk, or do not recognize when they have completed a sentence. As early as first grade, young children can “produce both simple and complex clausal constructions in their writing” (Berninger et al. 2010, p. 153). When required most first graders can produce a simple grammatically correct sentence.

Although development is not homogenous, as students get older the length of their sentences increase in writing, but do not necessarily increase in speech (O’Donnell 1967). In addition, students gain more syntactical control over their writing (O’Donnell 1967; Berninger et al. 2010), and make greater use of passive verbs (Myhill 2009). Older students are able to create increasingly more complex sentences for a variety of purposes, such as narrative, descriptive, or persuasive writing (Beers and Nagy 2011; Myhill 2009). Over time, grammar improves, and students are able to add details and include subordinate clauses. Older students are less likely to

produce sentence fragments or run-on sentences. Evidence also suggests that girls write longer sentences than boys (O’Donnell 1967). Studies analyzing sentence construction of individuals with ASD show mixed results. Some studies reveal that the syntactical abilities of individuals with ASD are relatively intact and are similar in length and complexity to their TD peers (Norbury and Bishop 2003). Whereas other studies show sentences written by individuals with ASD lack complexity (Myles et al. 2003). Some individuals with ASD show errors in referencing, e.g., pronouns do not agree with subjects and objects. This results in ambiguity (Norbury and Bishop 2003). It is not clear if this type of error is linked to social-communication deficits in individuals with ASD or to the age of the individuals.

Content

The content of written expression can be examined from two perspectives, the product and the process (Berninger et al. 1996). Production of text follows a developmental pattern. Young children first produce pictures without text, then representations of letters, true letters, sentences and coherent pieces of writing. Students can distinguish between different types of text (Beers and Nagy 2011), which include narrative, descriptive, informative, and persuasive forms of writing at a young age. Within each type there are conventions related to organization and language choice, e.g., a persuasive letter written to a legislator would look very different from a flyer on persuading readers not to take up smoking.

Developmentally, children progress from writing unrelated sentences, to listing or chaining associated ideas, to creating a coherent, organized text. This organization is evident in paragraphs in which details are organized around a local topic, and in longer pieces in which local topics are organized around a global topic (Berninger et al. 1996). Desirable features of an organized narrative include setting, problem, sequenced events, and climax, whereas a hierarchical structure developed around topic sentences is preferred in expository texts. Less skilled writers tend to use a knowledge-telling process in which relevant content is listed, whereas more mature writers transform this information in coherent, organized text.

The process of writing refers to the functions of planning, organization, and revising. It appears that executive functions contribute to the writing process. Executive control is a term used to describe functions related to planning, organization, and cognitive flexibility that enable an individual to obtain a particular goal. In writing, executive functions may govern processes in several ways: the retrieval of linguistic information for encoding, the generation of ideas, planning, and the monitoring of behaviors needed to produce text (Altemeier et al. 2008). Atypical executive function is often evident in individuals with ASD (Kenworthy et al.

2008). Impairments are most notable in working memory and planning tasks (Chen et al. 2016), and appear to be related to an individual's IQ, age and the complexity of the tasks presented. Context may also play a role in executive functioning, as individuals with ASD tend to perform better in highly-structured settings (Kenworthy et al. 2008). Notably, working memory deficits can impact the ability of individuals with ASD to retell a story, a task which requires the retrieval of relevant vocabulary, syntactical structure, and sequencing of events (Gabig 2008).

Oral Language and Early Literacy Experiences

It is also important to consider possible correlations between oral language skills and early literacy experiences on the writing performance of individuals with ASD. For individuals with specific language impairment (SLI), oral language measures on vocabulary and spelling have been found to be a significant predictor of compositional quality (Dockrell and Connelly 2015). Processes associated with spelling (phonological, orthographical, morphological), learned through oral language may be affected in individuals with SLI. However, Browder et al. (2009) found that the correlation between phonemic awareness skills and becoming a successful reader is not consistent in learners who display deficits in oral language ability. Instead, impairments in oral language processing may impact the development of word meaning for learners with ASD (McIntyre et al. 2017), which in turn may impact text generation.

Language experiences may account for some of the variations found in the performance of individuals with ASD. Although alphabet knowledge is equivalent to preschool peers (Westerveld et al. 2016), parents reported that their children with ASD did not appear to understand why people read and write (Lanter et al. 2012). Dynia, Lawton, Logan, and Justice (2014) found differences in the home literacy environment of children with ASD, with less frequent opportunities to participate in shared reading possibly explained by children not reciprocating an interest in literacy related opportunities.

In a study comparing oral language problems in individuals with language impairments and ASD, Dockrell, Ricketts, Charman, and Lindsay (2014) found many participants with ASD were not able to complete a writing task, possibly because they did not understand the task demands or were unable to generate ideas. Participants who did not produce text were significantly impaired on the Social Responsiveness Scale, a measure of the severity of social-communication impairments in autism. Evidence suggests that many individuals with ASD lack theory of mind (Baron-Cohen et al. 1985), meaning that they have difficulty interpreting the mental states of others. It has been suggested that a lack of theory of mind may contribute to impairments

in understanding human relationships (Brown and Klein 2011). Writing tasks often require the writer to understand the perspective of others, for example to recognize the view of the reader, the task demands, or multiple viewpoints. It is not known how the defining social-communication impairments of individuals with ASD might influence writing performance.

Need for Study

Current research has provided educators with a clear understanding of patterns of reading ability in individuals with ASD (Brown et al. 2013; Huemer and Mann 2010; Nation et al. 2006), however less is known about the written expression of individuals with ASD. A limited number of literature reviews related to individuals with ASD and writing were identified (Kushki et al. 2011; Pennington and Delano 2012; Westerveld et al. 2016). No reviews were identified considering the overall writing ability of individuals with ASD compared to their typically developing peers. Kushki et al. (2011) conducted a review of handwriting difficulties and students with ASD. Pennington and Delano (2012) conducted a review of instructional strategies for teaching writing to students with ASD, and Westerveld et al. (2016) conducted a review of the emergent literacy skills of preschoolers with ASD. Research synthesis is a systematic approach to present the current state of knowledge, and to highlight important issues that research has left uncovered (Cooper 2016). As such, the present study emerges as the first meta-analysis of the writing ability of individuals with ASD. Wang et al. (2013) note that it is common to select one of two approaches, meta-analysis or quality examination checks, when conducting a synthesis of the research. The present synthesis utilizes both approaches. A meta-analysis was conducted using quantitative procedures and statistical analysis to synthesize the prior research. Additionally, quality indicators were applied to studies meeting inclusion criteria as part of the analysis. The purpose of this study was through the examination of existing research, to compare the written expression of individuals with ASD and their typically developing (TD) peers.

The research questions under consideration follow: Are there differences between individuals with ASD and TD individuals in written expression? If so, in what components are there differences? Since many individuals with ASD exhibit grapho-motor deficits, it was hypothesized that they would have significantly lower scores in measures of handwriting. Due to cognitive differences affecting executive function and social understanding established in prior research, it was hypothesized that differences in the writing performance of individuals with ASD would emerge in components related to sustained and connected language use and fluency (e.g., content, length), and not in areas that are

rule-governed (e.g., sentence construction), or in which rote memory could be more readily utilized (e.g., spelling).

Method

Selection of Studies

Several strategies were employed to ensure a comprehensive search of studies (Cooper 2016). An electronic search of four databases, ERIC, Medline, PsychINFO, and SAGE yielded 70 articles. Databases were searched using all combinations of the following keywords, *autism spectrum disorders*, *autism*, *writing*, *writing skills* and *writing development*, to locate studies published between January 1985 and December 2015. Using the same search terms on Google Scholar yielded an additional eight articles. Issues of the five most recent years of the following journals were hand-searched for relevant articles: *Autism*, *Focus on Autism and Developmental Disabilities*, *Journal of Autism and Developmental Disorders*. These journals were selected for their impact ranking, except in the case of *Focus on Autism and Developmental Disabilities*, which was selected for its educational focus. Next, references from studies meeting inclusion criteria were reviewed to identify any additional studies meeting the inclusionary criteria. Finally, references from previous reviews were checked for relevant articles (Kushki et al. 2011; Westerveld et al. 2016). The researchers examined abstracts and methodologies from 82 studies, and identified 20% of the articles ($n = 16$) as appropriate for full review. Abstracts and methods were reviewed against the inclusion and exclusion criteria by both authors independently. Using the point-by-point approach (Kazdin 1982) inter-rater agreement was calculated at 87.5%. For the remaining studies, agreement was reached on whether or not the study should be included in the review. Two studies (Goldman 2008; Losh and Capps 2003) assessed transcribed oral records of participants' stories, which did not meet the inclusion criteria. One study (Mayes and Calhoun 2007) could not be included in this meta-analysis as relevant data were not obtainable. A total of 13 studies were included within the meta-analysis.

Inclusion and Exclusion Criteria

Studies were selected for this meta-analysis using the following predetermined criteria: (a) the study was published in English in a peer-reviewed journal between 2000 and 2015; (b) the participants in the study were individuals with ASD were diagnosed with either autism, ASD, Asperger's syndrome, pervasive developmental disorder, or pervasive developmental disorder-not otherwise specified; (c) the study included written expression as a dependent variable. Measures of written expression were derived from samples

in which participants produced connected meaningful text, such as spelling, sentences, or paragraphs. Text could be either handwritten or typed; (d) the study had a research design which included at least two comparison groups, one comprised of individuals with ASD and one comprised of TD individuals; (e) data on each group could be parsed out for analysis, i.e., means, standard deviations, and sample sizes could be obtained. If mean scores were not reported in the article, the corresponding author was contacted to request means and standard deviations. Exclusionary criteria were also applied. Studies in which participants with ASD could not be parsed out from the sample were excluded. Studies which included text dictated to a scribe were also excluded.

Coding Procedures

Attribute coding was used primarily for obtaining information from the studies, and the frequencies of each attribute were calculated for a descriptive summary. Attribute coding enabled the researchers to categorize the studies according to their dependent variables. Each study was coded for five descriptive attributes: (a) focus of journal, (b) number of participants, (c) age of participants, (d) inclusion criteria, and, (e) dependent variables.

Quality indicators were developed to assess the quality of the studies. A coding system based on quality indicators was developed by the researchers using procedures described by Cooper (2016). Although no interventions were administered, all the studies were experimental or quasi-experimental in that they tested the causal effects of an independent variable on a dependent variable, i.e., considered if a diagnosis of ASD had an effect on written expression. Quality indicators were based on the recommendations of Gersten et al. (2005). The 15 quality indicators (defined in Table 1) broadly fit into three categories: (a) fidelity in procedures for selecting participants, (b) fidelity of implementation and (c) fidelity of data analysis. The first author coded all 13 articles selected for the study. A random selection of half the articles were coded a second time by either a graduate student or the second author to establish reliability, which was calculated as 86% exact agreement. Discrepancies in scores were discussed and articles rescored for 100% exact agreement.

Analysis

Only six studies provided composite scores for any component of writing, three provided composite scores on handwriting (Fuentes et al. 2009, 2010; Hellinckx et al. 2013), one provided composite scores on mechanics (Brown et al. 2014), and three provided composite scores on content (Brown et al. 2014; Brown and Klein 2011; Myles et al. 2003). Across all the studies there were 98 measurements

Table 1 Definitions for study attributes and quality indicators

Attributes	Definition
Focus of journal	Autism spectrum disorder, neurology, occupational therapy, language
Number of participants	Total number of participants in each condition
Age of participants	Mean age in years of participants
Inclusion criteria	Diagnosis only, IQ, age, exclusionary criteria
Dependent variables	Length, handwriting, spelling, sentence construction, content
Quality indicator	Definition
Participants	
Comparable groups	Groups were same size, and/or measures were taken to ensure groups were comparable
Age	The mean age of the participants in the comparison groups was not significantly different
Eligibility	Diagnostic information was used to determine participants met criteria for inclusion in the study
Treatment fidelity	Participants across groups were treated consistently and tested in similar settings
Random selection	Participants randomly selected from a larger representative group
Implementation	
Testers	Study indicates whether testers were qualified and/or trained
Number	Participants in groups received a comparable number of tests
Measures	Study can be replicated with same outcome measures
Reliability	Standardized tests were used or interrater reliability was reported
Validity	Evidence provided that measures were valid
Data analysis	
Techniques	Analysis linked to research questions
Relationship between variables	Interaction effects and/or correlation scores reported to indicate whether or not there is a relationship between the variables
Data reporting	Adequate data reported, including gender and age of participants
Directionality	Direction of outcome measures given if applicable
Power analysis	Power analysis conducted to determine sample size

given. Four studies included variables on content, spelling and sentence construction (Brown et al. 2014; Brown and Klein 2011; Myles et al. 2003; Troyb et al. 2014). Since it was not possible to create a meaningful composite score from the data provided, it was decided not to create a summary effect for all the variables across all the studies.

Using EXCEL, variables were coded, first according to the writing component (handwriting, spelling, sentence construction, or content), and then according to their measurement. Many studies included measures of length, so this category was created for the purposes of analysis. Variables measuring the number of letters, words, t-units, morphemes, sentences, or clauses were grouped under length. Definitions and frequency of variables are listed in Table 3. Brown and Klein (2011) collected two sets of data from participants, one set for narrative writing and one set for expository writing. The data on narrative writing were included in this meta-analysis as it most closely resembles writing tasks in other studies (Myles et al. 2003; Troyb et al. 2014). In addition, measurements that were derived from other measurements (e.g., percentage of t-units or mean number of letters per word) were excluded from the meta-analysis. A total of 35 variables were included in the meta-analysis.

Analysis on each component of writing was also conducted to determine whether differences existed in the written expression of participants with ASD and TD participants, in certain components of writing. Corresponding variables which could be identified in four or more studies were analyzed using the Comprehensive Meta-Analysis (CMA) computer program (Borenstein, Hedges, Higgins, & Rothstein, 2005). Effect sizes on measures comparing outcomes between participants with ASD and the control groups were calculated using Hedges *g*. Hedges *g* was chosen because sample sizes in the studies tended to be small. Hedges *g* is adjusted for bias, which is appropriate given the variability in the measures used across the settings. Variables did not always have clear directionality, e.g., larger letter sizes may not affect legibility and may not constitute a negative value. For the purposes of calculating effect sizes directionality was assigned as in the original research. Therefore large letter size and simple sentences were considered to be a less desirable outcome, and complex sentences were considered to be more desirable.

Even though there was not much variability in the participants within studies, there was variability across studies, most notably age. Two studies matched and ranked

participants for analysis (Cartmill et al. 2009; Lanter et al. 2013). In seven studies, independent t-tests were used, however participants in each group had to meet similar criteria to be included in the study (Beverdors et al. 2001; Fuentes et al. 2009, 2010; Hellinckx et al. 2013; Myles et al. 2003; Troyb et al. 2014; Tyson et al. 2014). Due to variability across studies, summary effect sizes were calculated on each group of variables using a random effects model. Heterogeneity was assessed using the Q test (Borenstein et al. 2005). I^2 and τ^2 are reported as a complement to the Q test.

All the studies were coded for quality indicators defined in Table 1. For each quality indicator a score of 1 was assigned if there was evidence in the publication that it was met. Each quality indicator was scored as 1 (met) or 0 (not met). If information regarding a specific quality indicator was not provided, a score of 0 was given. Quality indicator points were added for a total quality score, resulting in a scale of 0–15. To determine if there was a relationship between the quality of the studies and the magnitude of the effect sizes, Pearson Product-Moment correlation coefficient was calculated using the quality indicator totals and effect sizes for each study.

Results

There were a total of 750 participants, aged 4–42 years included in the 13 studies used for this meta-analysis (ASD = 354, TD = 360, other = 102). Data on gender were not available for 130 participants. The ratio of male to female participants in the remaining participants was 7:3. For the participants with ASD, clinical diagnosis was required in eight of the studies, referral or parental report for the others (Table 2). Eight of the studies required participants to meet a minimum IQ score, and nine excluded participants with other conditions that might impact results e.g., visual impairment, neurological disorders. Studies were published in peer-reviewed articles that focused on ASD ($n = 8$), neurology ($n = 2$), occupational therapy ($n = 3$), and language ($n = 1$). Handwritten samples were used in 11 of the studies, and typed responses were used in two studies (Brown et al. 2014; Brown and Klein 2011). European characters were used in all writing samples.

Component Comparisons

Length

Four studies included 16 different measures of length (Brown et al. 2014; Brown and Klein 2011; Myles et al. 2003; Troyb et al. 2014). All four studies included a measure of the total number of words produced in a spontaneous writing sample. Number of words written was

significantly lower for individuals with ASD $g = 0.671$ ($Z = 4.070$, $SE = 0.165$, $p < .01$). Participants with ASD wrote significantly less words than typically developing participants. There were not enough measures of length in t-units, clauses, or morphemes for meta-analysis. The review of individual studies shows that comparisons on t-units, clauses and morphemes between participants with ASD and typically developing participants were similar to results comparing the number of words written. However, the number of sentences participants with ASD wrote was not always significantly different (Myles et al. 2003; Troyb et al. 2014).

Handwriting

This analysis yielded 28 variables across seven studies on handwriting (Alaniz et al. 2015; Beverdors et al. 2001; Cartmill et al. 2009; Fuentes et al. 2009, 2010; Hellinckx et al. 2013; Lanter et al. 2013). There were enough measures on legibility, letter formation, size, and speed to warrant statistical analysis. Results are presented in Table 4. Scores for lower case letters were selected if there was a choice. Summary effects also showed that participants with ASD had significantly lower scores on legibility ratings, $g = 0.664$ ($Z = 3.889$, $SE = 0.171$, $p < .01$). In addition, summary effects showed that handwriting was also significantly larger for participants with ASD $g = 0.745$ ($Z = 4.811$, $SE = 0.155$, $p < .01$), and measures of speed revealed that participants with ASD wrote significantly slower than TD participants $g = 0.651$ ($Z = 4.947$, $SE = 0.132$, $p < .01$). Summary effects did not show any significant differences on scores related to letter formation. Three studies measured alignment and spacing (Fuentes et al. 2009, 2010; Hellinckx et al. 2013) and two measured consistency (Cartmill et al. 2009; Hellinckx et al. 2013). Results were mixed.

Spelling

Summary effects were calculated using data from six variables across six studies (Asberg et al. 2010; Brown et al. 2014; Brown and Klein 2011; Cartmill et al. 2009; Myles et al. 2003). In three studies (Asberg et al. 2010; Cartmill et al. 2009; Myles et al. 2003), participants were required to spell a word given in context. In the two studies conducted by Brown et al. (2014) error rates were calculated using a writing sample, and in Brown and Klein (2011) a spelling score was calculated using a writing sample. Results showed that spelling scores for participants with ASD were significantly lower than scores for TD participants, $g = 0.283$ ($Z = 2.140$, $SE = 0.132$, $p = .032$).

Table 2 Attributes and quality indicators scores of each study

Author	Focus of journal ^a	Number of participants	Mean (SD) age of participants in years	ASD criteria ^b	Dependent variables	Quality indicator score
Alaniz et al. (2015)	Occupational therapy	ASD = 27 TD = 24	6.75 (1.75)	Community diagnosis	Handwriting	11
Asberg et al. (2010)	Language	ASD = 20 ADHD = 36 TD = 54	ASD = 11.8 (2.7) ADHD = 13.0 (2.4) TD = 12.5 (2.6)	Clinical diagnosis IQ > 70	Spelling	13
Beversdorf et al. (2001)	Autism	ASD = 10 TD = 13	ASD = 30.8 (9.3) TD = 30.6 (12.8)	ADI-R IQ ≥ 85	Handwriting	8
Brown and Klein (2011)	Autism	ASD = 16 TD = 16	ASD = 25.75 (7.9) TD = 26.56 (7.0)	Self-report of ASD IQ ≥ 85	Content Length Spelling Sentence construction	10
Brown et al. (2014)	Autism	ASD = 25 TD = 22	ASD = 12.91 (2.15) TD = 13.09 (2.5)	Documented diagnosis IQ ≥ 80	Content Length Spelling Sentence construction	12
Cartmill et al. (2009)	Autism	ASD = 28 TD = 28	ASD = 8.5 (0.38) TD = 8.38 (0.32)	Gilliam Autism Rating Scale	Handwriting Spelling	14
Fuentes et al. (2009)	Neurology	12 ASD = 14 TD = 14	ASD = 10.2 (1.9) TD = 11.1 (1.3)	Clinical diagnosis IQ ≥ 80	Handwriting	13
Fuentes et al. (2010)	Neurology	ASD = 12 TD =	ASD = 14.4 (1.4) TD = 13.8 (1.2)	Clinical diagnosis	Handwriting	12
Hellinckx et al. (2013)	Autism	ASD = 69 TD = 61	ASD = 10.49 (1.49) TD = 10.33 (1.23)	Clinical diagnosis IQ ≥ 80	Handwriting	13
Lanter et al., (2013)	Autism	ASD = 32 TD = 32	ASD = 5.62 (1.22) TD = 2.75 (0.66)	Community diagnosis	Handwriting	11
Myles et al. (2003)	Autism	ASD = 16 TD = 16	ASD = 11.58 TD = 11.5	Clinical diagnosis	Content Length Spelling Sentence construction	13
Troyb et al. (2014)	Autism	ASD = 41 OO = 41 TD = 34	ASD = 13.81 (2.67) OO = 12.91 (3.47) TD = 13.87 (2.58)	Documented diagnosis	Content Length Sentence construction	12
Tyson et al. (2014)	Autism	ASD = 44 OO = 34 TD = 34	ASD = 13.9 (2.7) OO = 12.8 (3.5) TD = 13.9 (2.6)	Clinical diagnosis IQ ≥ 77	Sentence construction	13

^aNo studies were found in journals focusing on psychology and neuroscience

^bInstrument is listed if diagnosis was confirmed by researchers, clinical diagnosis = medical records, community diagnosis = school or medical records, documented = source of diagnosis not specified

Sentence Construction

Five studies (Brown et al. 2014; Brown and Klein 2011; Myles et al. 2003; Troyb et al. 2014; Tyson et al. 2014) included 20 measurements on variables related to syntax (Table 3). All four studies included a measure on mechanics. Brown et al., (2014) included a measure of grammatically correct sentences, Brown and Klein (2011) counted grammatical errors, and two studies reported TOWL scores on contextual language, a measure of vocabulary, sentence

structure, and grammar (Myles et al. 2003; Troyb et al. 2014). These four scores were considered to be similar enough for meta-analysis. No significant difference was found in the summary effect scores on mechanics between participants with ASD and TD participants.

Two studies (Brown et al. 2014; Brown and Klein 2011) measured the percentage of either sentences or t-units, which made reference to previously written material. Only in regard to expository text were rubric scores significantly lower for participants with ASD (Brown and Klein 2011).

Table 3 Definition and frequency of variables

Component	Variable	Definition	Frequency	Number of studies
Length	Length	Total number of words, t-units sentences, or clauses	7	4
Handwriting	Legibility	Letters must be present, recognizable and have all parts complete	4	4
	Letter formation	Accurate formation of letters with no additional strokes	5	5
	Alignment	Letters are aligned with baseline	3	3
	Size	Letter height	5	5
	Spacing	Letters do not touch and there is space between words	3	3
	Consistency	Formation is consistent and letters are connected	2	2
	Speed	Number of letters written in a given time frame	4	4
	Spelling	Ability to spell a target word presented in a sentence or number of errors in a writing sample	5	5
Sentence construction	Complexity	Frequency of different types of sentences (simple, compound, complex), or clausal density (total number of clauses divided by the total number of t-units)	5	2
	Mechanics	Grammar, capitalization and punctuation	7	4
	Referents	Percentage of sentences which made reference to previous sentences	4	3
	Syntax	Frequency of sentences formulated correctly with prompts, run-on sentences or sentence fragments	4	2
Content	Topic	Ability to stay on topic and include relevant information	4	2
	Structure	Writing follow conventions of genre with clear paragraphs	4	4
	Vocabulary	Writing contains either multi-syllabic words, a variety of words, or contextual language makes sense	5	2
	Style	Respectful and appropriate tone	1	1
	Pragmatics	Frequency of words or sentences which convey emotions and/or thoughts or a character or the author. Frequency of words or sentences referring to causes or motives	10	2

Two studies (Brown et al. 2014; Myles et al. 2003) included measures of complexity. Significant differences were found on clausal density (Brown et al. 2014), and the percentage of compound sentences (Myles et al. 2003) suggesting that TD participants wrote more complex sentences than participants with ASD. However no significant differences were found on other measures. In regard to syntax, Tyson et al. (2014) found that participants with ASD scored significantly lower on formulating sentences. However, Myles et al. (2003) found no significant differences between groups in the % of sentence fragments or run-on sentences.

Content

Four studies (Brown et al. 2014; Brown and Klein 2011; Myles et al. 2003; Troyb et al. 2014) included 25 measures of content. Two studies included rubric scores related to the quality of writing within a particular genre, narrative structure (Brown and Klein 2011), and persuasive writing (Brown et al. 2014) and two studies (Myles et al. 2003; Troyb et al. 2014) reported TOWL scores on story construction. These scores were considered to be similar enough for meta-analysis on structure. Scores were found to be significantly lower for participants with ASD than their TD counterparts

$g = 0.550$ ($Z = 3.564$, $SE = 0.154$, $p = .001$). These same four studies also collected data on various measures related to style, vocabulary, and pragmatics. Brown and Klein (2011) found that participants with ASD scored significantly lower on rubric scores on the degree to which text fit with an assigned topic. No other significant differences were found.

Heterogeneity

The Q statistic (Table 4) was significant for length and letter formation. For letter formation, $I^2 = 95.346$, indicating that a high proportion of the observed variance reflects real differences in effect size. For many of the variables, legibility, speed, spelling sentence construction and structure Tau^2 set to zero. The Q score was less than the degrees of freedom, as such it was not possible to calculate I^2 or Tau^2 . It is not clear whether or not effect sizes are due to real differences in the populations or due to random sampling error. Except for the variables of length and letter formation, the Q score is insignificant, meaning the null hypothesis, which asserts the studies share a common effect size, cannot be rejected. Q scores could be due to low power resulting from a small number of studies.

Table 4 Summary effects and Q scores for variables

Variable	Studies	Hedges <i>g</i>	<i>z</i> -score	<i>SE</i>	<i>p</i>	<i>Q</i> score	<i>I</i> ²	<i>Tau</i> ²	<i>p</i>
Length									
Length in words	Brown et al. (2014) Brown & Klein (2011) Myles et al. (2003) Troyb et al. (2014)	0.671	4.070	0.165	<0.01*	3.328	9.853	0.011	<0.01*
Handwriting									
Legibility	Alaniz et al. (2015) Cartmill et al. (2009) Fuentes et al. (2009) Fuentes et al. (2010)	0.664	3.889	0.171	<0.01*	0.965	0.00	0.00	0.81
Letter formation	Cartmill et al. (2009) Fuentes et al. (2009) Fuentes et al. (2010) Hellincks et al. (2013) Lanter et al. (2012, 2013)	0.619	1.024	0.605	0.306	85.943	95.346	1.727	<0.01*
Size	Beversdorf et al. (2001) Cartmill et al. (2009) Fuentes et al. (2009) Fuentes et al. (2010) Hellincks et al. (2013)	0.745	4.811	0.155	<0.01*	5.205	23.144	0.028	0.267
Speed	Cartmill et al. (2009) Fuentes et al. (2009) Fuentes et al. (2010) Hellincks et al. (2013)	0.651	4.947	0.132	<0.01*	1.049	0.00	0.00	0.789
Spelling									
Spelling	Asberg et al. (2010) Brown et al. (2014) Brown & Klein (2011) Cartmill et al. Myles et al. (2003)	0.283	2.140	0.132	0.032*	3.694	0.00	0.00	0.449
Sentence construction									
Mechanics/grammar	Brown et al. (2014) Brown & Klein (2011) Myles et al. (2003) Troyb et al. (2014)	0.289	1.899	0.152	0.058	2.964	0.00	0.00	0.397
Content									
Structure	Brown et al. (2014) Brown and Klein (2011) Myles et al. (2003) Troyb et al. (2014)	0.550	3.564	0.154	0.001*	2.914	0.00	0.00	0.405

**p* < .05

Quality of Studies

The results for the 15 quality indicators are listed in Table 2. Studies met a range of 8–14 indicators, with no studies meeting all 15 quality indicators. There was variability in quality across all studies with < 1% of the investigations meeting 14 indicators, 30.77% of the investigations meeting 13 indicators (mode = 13), 23.07% of investigations meeting 11 and 23.07% meeting 12 indicators, and the remaining investigations meeting 10 or less indicators. Across all studies, the quality was weakest in

the indicators of randomization and power analysis. Just 15.38% of investigations selected participants randomly from a larger group with the majority utilizing samples of convenience. Only 15.38% of studies reported conducting a power analysis to determine sample size. Also in relation to data analysis, 69.23% of investigations reported interaction effects and/or correlations. Quality was highest across studies (> 90%) in the indicators of age, treatment fidelity, number of tests, replicability of measures, reliability, data techniques, adequacy of data reported, and directionality of outcome measures.

Effect Sizes

Effect sizes ranged from 0.28 to 0.75 across the components of size (0.75), length (0.67), legibility (0.66), speed (0.65), letter formation (0.62), structure (0.55), sentence construction (0.29) and spelling (0.28). Correlations were conducted to investigate the relationship of study quality and the effect sizes of the six writing components identified with significant differences between individuals with ASD and TD individuals (size, length, legibility, speed, structure, and spelling). There was little relation found between the quality indicator scores and the magnitude of study effect sizes. Of note, while not significant, negative correlations were found between effect size and study quality in four of the six components: size, length, legibility, and structure.

Discussion

It was hypothesized that the writing performance of individuals with ASD would be significantly different from their typically developing peers in measures of handwriting, and in the components that required sustained and connected language use, e.g., length, and content. Conversely, significant differences would not be found in spelling or sentence construction. This study compared the performance in written expression of individuals with ASD and their TD counterparts by conducting a meta-analysis using data from previously published studies. All thirteen studies selected were used in the meta-analysis. There were significant differences in performance in six of the eight components of writing analyzed; (a) length, (b) legibility, (c) size, (d) speed, (e) spelling, and (f) structure. Results confirmed the hypothesis that individuals with ASD would have lower scores in handwriting. The handwriting of ASD participants was bigger in size and scored lower on measures of legibility. Contrary to the hypothesis, spelling scores were significantly lower for participants with ASD. Apart from letter formation, individuals with ASD scored lower in measures related to transcription skills (handwriting and spelling). This could be due to grapho-motor differences in individuals with ASD. However, results may be skewed due to the high proportion of boys with ASD, thereby reflecting established patterns of handwriting problems in boys (Berninger et al. 1997; Graham et al. 1998). Although phonological awareness is a predictor of spelling ability (Garcia et al. 2010), other explanations for lower spelling scores in individuals with ASD must be considered. Other elements of oral language e.g., vocabulary development (McIntyre et al. 2017) and morphological processes, need to be examined further in individuals with ASD to secure an understanding of their spelling development.

Given the relationship between transcription skills and content (Berninger et al. 1997), findings on the components of length and content were unsurprising. Participants with ASD wrote less and at a slower rate than TD participants. On rubric indicators measuring the structure of writing, participants with ASD scored significantly lower. The findings of this study suggest individuals with ASD experience difficulties with written expression, and those difficulties are analogous to those experienced by other individuals identified by researchers as poor writers (Berninger 1999; Graham et al. 2008; Graham and Harris 2006). As hypothesized there were no significant differences found in the component of sentence construction, which suggests that like their typically developing peers, individuals with ASD are benefitting from existing educational interventions in this area.

There were no significant correlations between quality indicator scores and the effect sizes (Hedges *g*) of measures included in the meta-analysis on those variables that were found to be significant. All the studies included in the present study were awarded a quality indicator score of 8 or above. It appears that the quality of the all the studies were sufficient and it is unlikely that variations in quality indicator scores affected the overall findings of the meta-analysis.

Limitations

The findings of this meta-analysis should be viewed with caution, as there are several limitations to the study. First, the sample size is small. Only 13 studies met criteria for inclusion, and although these studies yielded a total of 98 different measurements, only 35 were used for meta-analysis. A small sample size can also impact effect sizes which were small to medium, yet a larger sample might not impact effect sizes. In nine studies, participants were matched for either gender, age or IQ to prevent sampling errors. In this context, the small to medium effect sizes suggest, despite difference in underlying etiology, writing performance in this subset of individuals is not too dissimilar to other students with learning difficulties, and their writing development may follow a similar trajectory to their typically developing peers.

A second issue was the range of measures. Studies reported data on different measures of the same concept, e.g., length, or included measures created from other measures e.g., percentages, thus making it difficult to find comparable measures. Insufficient data were available to analyze measures in vocabulary and pragmatics, thus resulting in a limited view of written expression by individuals with ASD. A third limitation is the variability of participants. Nearly all the studies included participants with a verifiable clinical diagnosis of ASD. IQ scores greater than 70 were listed in the inclusion criteria of seven studies, as such results may be skewed. Strict criteria regarding IQ and oral language abilities may result in samples not reflecting the variability

in IQ (Griswold et al. 2002), vocabulary (Barnhill et al. 2000), and oral language (Westerveld et al. 2016) found in the overall population of individuals with ASD. Since the incidence of ASD is approximately 1 in 68 (CDC 2014), random selection of participants from a larger group could be considered in future studies. Across studies the ages of participants varied considerably. Measures were sometimes related to the age of participant e.g., in Lanter et al. (2013) emerging literacy skills were measured, whereas in Brown and Klein's analysis (2011), adult participants wrote a complete piece of persuasive prose. A fourth limitation is that the studies grouped by writing component did not always use the same assessment. Measures were not always identical, e.g., in the measurement of speed Fuentes et al., (2009) measured rate of speed over 2.5. min, whereas Hellinckx et al. (2013) measured rate of speed over 5 min. It is not clear if the length of time spent writing impacted participants' performance in speed or other variables. Likewise it is impossible to determine how the lack of correspondence in measures impacted overall results.

Fifth, in considering a relationship between the magnitude of effect size and quality of studies resulting in a significant difference between individuals with ASD and TD individuals, the range of effect sizes was limited in variability with five of the six falling between 0.6 and 0.7, potentially limiting the significance of correlations with quality indicators. Finally, a low quality indicator score could be considered a limitation. Many quality indicators pertain to ideal sampling methods (e.g., random sampling and power analysis to determine sample size) which are difficult to attain with a population of individuals with ASD. Lack of data on gender or IQ prohibited analysis of subgroups.

Implications for Research

The results of this meta-analysis suggest that individuals with ASD struggle with written expression. In many respects individuals with ASD exhibit traits similar to other individuals with learning problems. It appears that difficulties in handwriting and spelling affect the overall quality of written composition. However, many questions remain. It is not clear whether individuals with ASD exhibit differences in vocabulary use or in the use of pragmatic language, such as that needed for character development. Further research also needs to be done to determine if individuals with ASD have strength in writing for a particular purpose. For example, do they perform better in expository writing tasks than in persuasive writing tasks?

Many of the studies included in this meta-analysis used standardized measures. However, few standardized measures exist for assessing different genres of writing. As these measures are developed, researchers should seek

to establish reliability and validity so that comparisons between and within populations can be reasonably made.

Although sampling methods may be limited by geographical area and availability when studying a population with ASD, this should not deter researchers from addressing other quality indicators. This meta-analysis did not calculate effect sizes for subgroups such as gender and age. As more data are collected, it is important researchers report data for these subgroups. This is necessary to determine, for example, how much of the variation in handwriting between individuals with ASD and typically developing individuals can be attributed to gender or to diagnosis.

Finally, it is also important that researchers consider the needs of individuals with ASD whose IQ scores are below 70. Does their development of written expression correspond with individuals with intellectual disabilities, or does it exhibit unique traits? Identifying specific deficits of individuals with ASD will enable researchers and educators to develop strategies to better support their academic needs.

Implications for Practice

This study found that individuals with ASD have difficulties with handwriting, spelling and the structure of writing. Patterns in performance are not dissimilar to other individuals who struggle with written expression. Current research is focused on whether existing strategies to improve written expression in students with learning difficulties will be effective for individuals with ASD (Pennington and Delano 2012). If the needs of individuals with ASD are similar to other individuals with learning problems, then educators can feel more confident with placing students with ASD in more inclusive environments to receive those same interventions.

Students with ASD are a heterogeneous group, as such educators need to recognize that not all their students' learning needs will be the same. Educators are advised to evaluate writing samples for length, legibility, size, speed, spelling, syntax, and structure using standardized measures, curriculum-based measurements and/or rubrics. Once student needs are identified, educators will then be better able to target accommodations and interventions. Educators also need to be aware that inherent characteristics of ASD may impact components of written expression (e.g., vocabulary, perspective taking in persuasive writing) which existing research does not sufficiently address. Educators should consider research-based practices designed for other populations to address learning needs in these areas. It is recommended the progress of each individual with ASD be closely monitored to determine the effectiveness of interventions.

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